DBMS – SQL QUERY IMPLEMENTATION FOR E-COMMERCE DATABASE

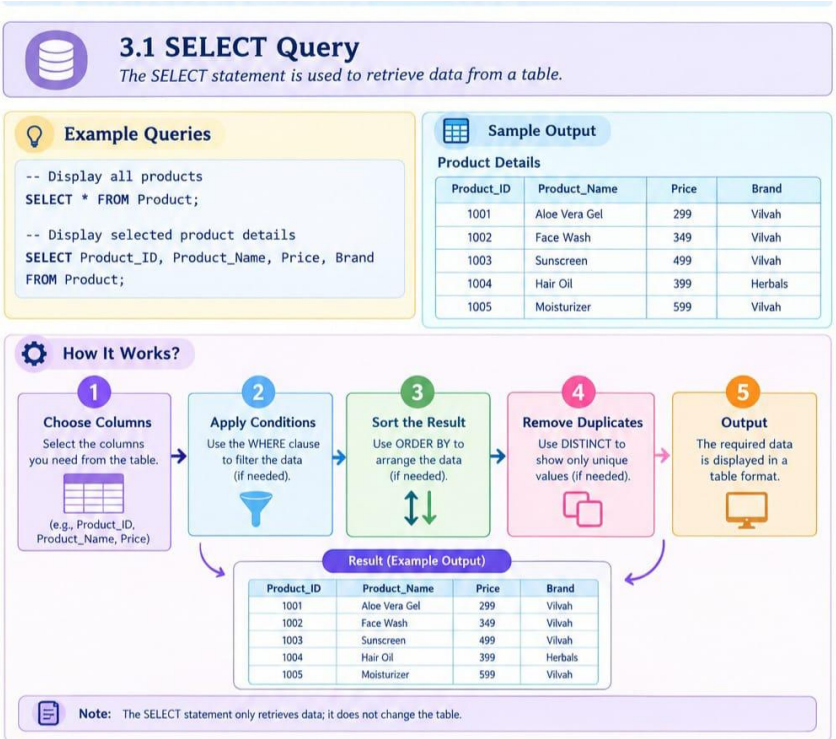
INTRODUCTION

SQL (Structured Query Language) is used to retrieve, manage, and analyze data stored in a database. In the Vilvah E-Commerce Database, SQL queries are used to display customer and product details, search products based on price and category, check product availability, and generate business reports. Commands such as SELECT, WHERE, ORDER BY, and DISTINCT help filter, sort, and organize data efficiently. These queries support effective management of products, customers, inventory, and orders in the Vilvah database.

Objectives

1. Perform SELECT, WHERE, ORDER BY, and DISTINCT queries.
2. Search products based on price, category, and availability.
3. Retrieve customer and product information.
4. Apply filtering conditions.
5. Generate basic business reports.

1. Basic SQL Queries



1.1 SELECT Query

Display all products

```
SQL> SELECT * FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID SUPPLIER_ID      PRICE WEIGHT
----------- ----------- ---------- --------------------
INGREDIENTS
--------------------------------------------------------------------------------
SKIN_TYPE
--------------------------------------------------
BRAND                                              EXPIRY_DA
-------------------------------------------------- --------
      1002

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID SUPPLIER_ID      PRICE WEIGHT
----------- ----------- ---------- --------------------
INGREDIENTS
--------------------------------------------------------------------------------
SKIN_TYPE
--------------------------------------------------
BRAND                                              EXPIRY_DA
-------------------------------------------------- --------
Rose Face Cream

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID SUPPLIER_ID      PRICE WEIGHT
----------- ----------- ---------- --------------------
INGREDIENTS
--------------------------------------------------------------------------------
SKIN_TYPE
--------------------------------------------------
BRAND                                              EXPIRY_DA
-------------------------------------------------- --------
          1         202        399 50g

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID SUPPLIER_ID      PRICE WEIGHT
----------- ----------- ---------- --------------------
INGREDIENTS
--------------------------------------------------------------------------------
SKIN_TYPE
--------------------------------------------------
BRAND                                              EXPIRY_DA
-------------------------------------------------- --------
Rose, Shea Butter

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID SUPPLIER_ID      PRICE WEIGHT
----------- ----------- ---------- --------------------
INGREDIENTS
--------------------------------------------------------------------------------
SKIN_TYPE
--------------------------------------------------
BRAND                                              EXPIRY_DA
-------------------------------------------------- --------
Dry Skin

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID SUPPLIER_ID      PRICE WEIGHT
----------- ----------- ---------- --------------------
INGREDIENTS
```

```
BRAND                                              EXPIRY_DA
-------------------------------------------------- ---------
          9         203        329 150ml

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID SUPPLIER_ID      PRICE WEIGHT
----------- ----------- ---------- --------------------
INGREDIENTS
--------------------------------------------------------------------------------
SKIN_TYPE
--------------------------------------------------
BRAND                                              EXPIRY_DA
-------------------------------------------------- ---------
Aloe Vera, Green Tea

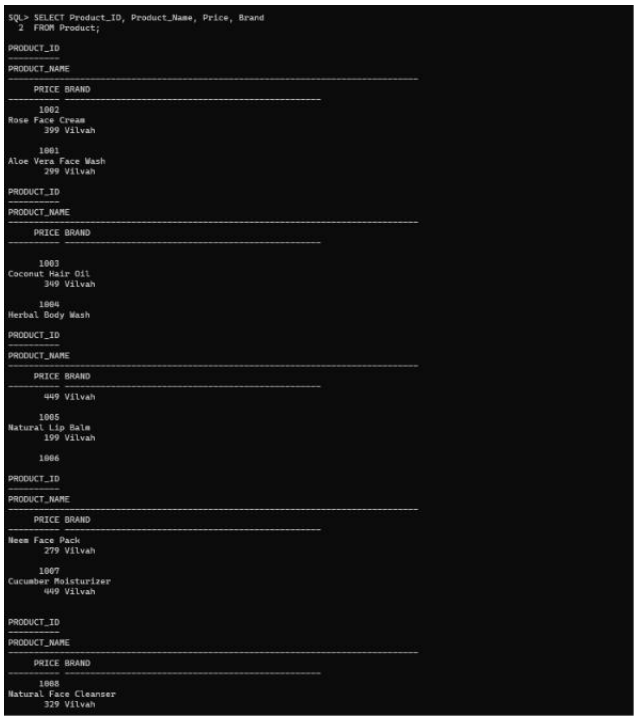
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID SUPPLIER_ID      PRICE WEIGHT
----------- ----------- ---------- --------------------
INGREDIENTS
--------------------------------------------------------------------------------
SKIN_TYPE
--------------------------------------------------
BRAND                                              EXPIRY_DA
-------------------------------------------------- ---------
Combination Skin

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID SUPPLIER_ID      PRICE WEIGHT
----------- ----------- ---------- --------------------
INGREDIENTS
--------------------------------------------------------------------------------
SKIN_TYPE
--------------------------------------------------
BRAND                                              EXPIRY_DA
-------------------------------------------------- ---------
Vilvah                                             12-DEC-27

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID SUPPLIER_ID      PRICE WEIGHT
----------- ----------- ---------- --------------------
INGREDIENTS
--------------------------------------------------------------------------------
SKIN_TYPE
--------------------------------------------------
BRAND                                              EXPIRY_DA
-------------------------------------------------- ---------


8 rows selected.
```

Display selected product details



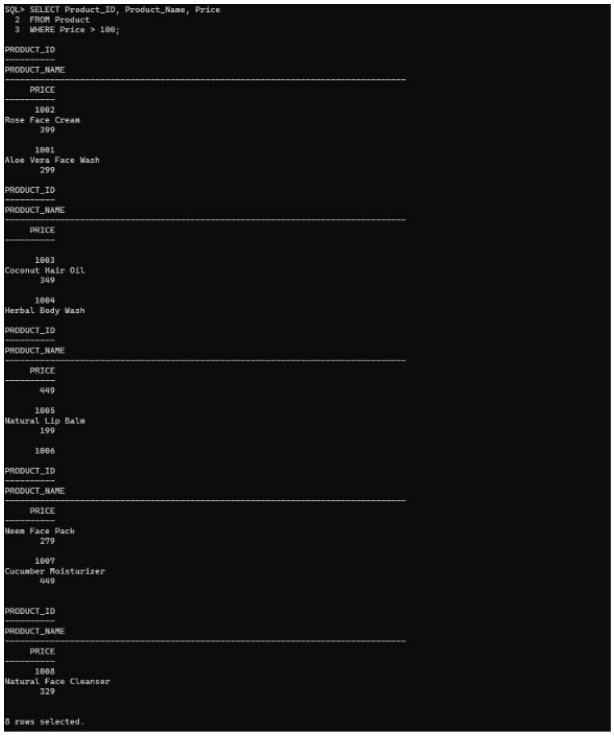
1.2 WHERE Clause

The WHERE clause is used to filter records based on a condition.

Find products with a price less than ₹1000



Find products with a price greater than ₹100



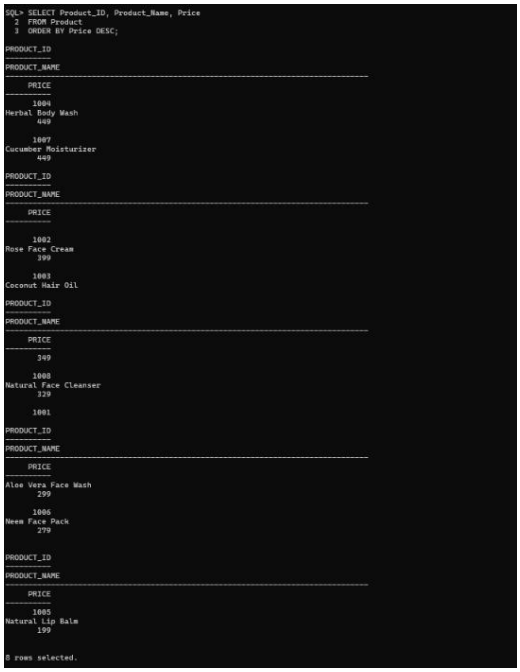
1.3 ORDER BY

The ORDER BY clause is used to sort query results.

Products from lowest to highest price



Products from highest to lowest price



1.4 DISTINCT

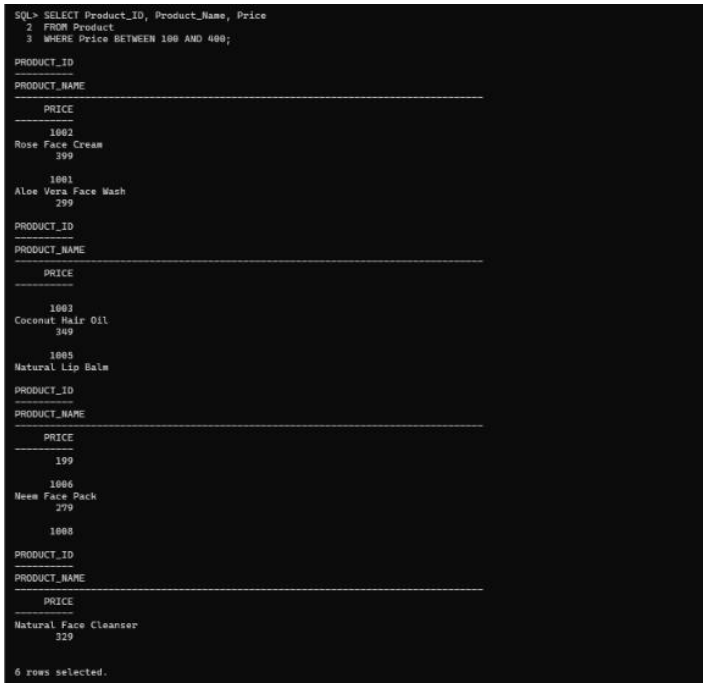
The DISTINCT keyword is used to display unique values without duplicates.

Display unique product categories



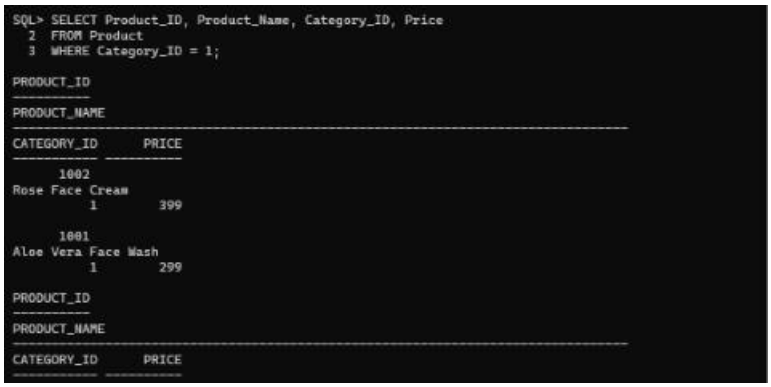
2. Search Products

2.1 Search Products Based on Price



This displays products whose price is between ₹100 and ₹400, including both limits.

2.2 Search Products Based on Category

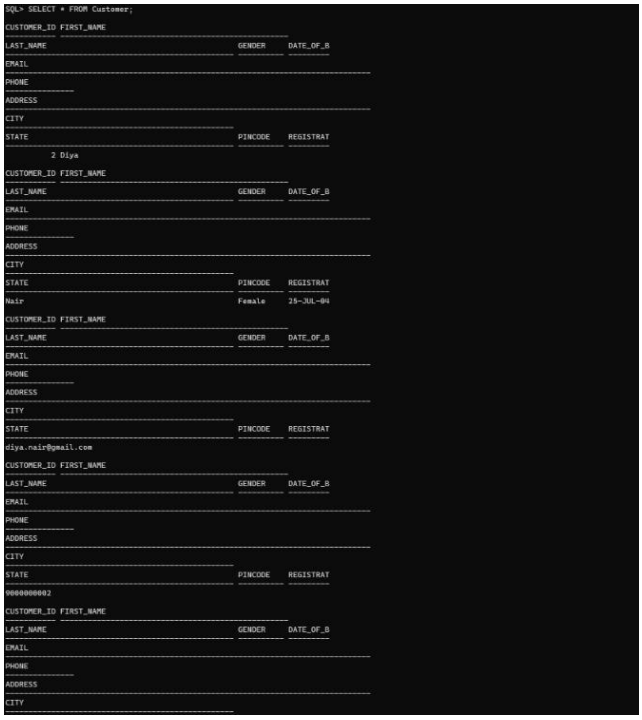


This displays products belonging to Category ID 1, which is Face Care in your Vilvah database.

3. Retrieve Customer Information

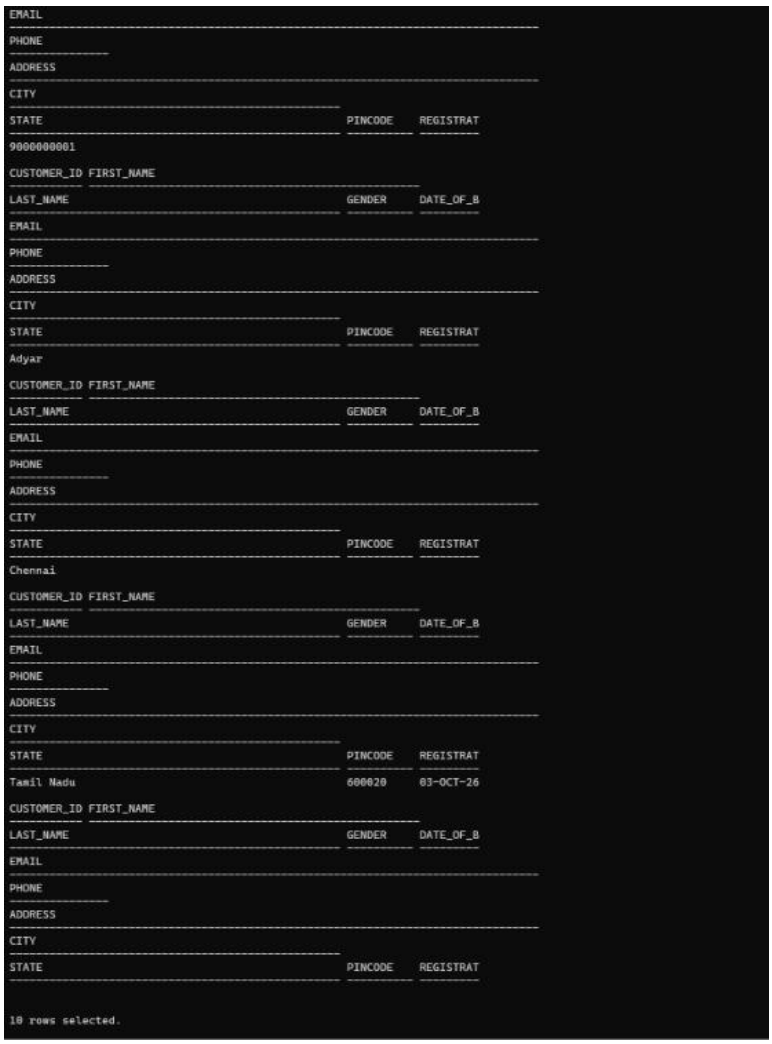
Display all customers







```
CUSTOMER_ID FIRST_NAME
LAST_NAME                                          GENDER     DATE_OF_B
EMAIL
PHONE
ADDRESS
CITY
STATE                                              PINCODE    REGISTRAT
9000000010
CUSTOMER_ID FIRST_NAME
LAST_NAME                                          GENDER     DATE_OF_B
EMAIL
PHONE
ADDRESS
CITY
STATE                                              PINCODE    REGISTRAT
Vastrapur
CUSTOMER_ID FIRST_NAME
LAST_NAME                                          GENDER     DATE_OF_B
EMAIL
PHONE
ADDRESS
CITY
STATE                                              PINCODE    REGISTRAT
Ahmedabad
CUSTOMER_ID FIRST_NAME
LAST_NAME                                          GENDER     DATE_OF_B
EMAIL
PHONE
ADDRESS
CITY
STATE                                              PINCODE    REGISTRAT
Gujarat                                            380015     03-OCT-26
CUSTOMER_ID FIRST_NAME
LAST_NAME                                          GENDER     DATE_OF_B
EMAIL
PHONE
ADDRESS
CITY
STATE                                              PINCODE    REGISTRAT
```



```
EMAIL
PHONE
ADDRESS
CITY
STATE                                              PINCODE    REGISTRAT
9000000001
CUSTOMER_ID FIRST_NAME
LAST_NAME                                          GENDER     DATE_OF_B
EMAIL
PHONE
ADDRESS
CITY
STATE                                              PINCODE    REGISTRAT
Adyar
CUSTOMER_ID FIRST_NAME
LAST_NAME                                          GENDER     DATE_OF_B
EMAIL
PHONE
ADDRESS
CITY
STATE                                              PINCODE    REGISTRAT
Chennai
CUSTOMER_ID FIRST_NAME
LAST_NAME                                          GENDER     DATE_OF_B
EMAIL
PHONE
ADDRESS
CITY
STATE                                              PINCODE    REGISTRAT
Tamil Nadu                                         600020     03-OCT-26
CUSTOMER_ID FIRST_NAME
LAST_NAME                                          GENDER     DATE_OF_B
EMAIL
PHONE
ADDRESS
CITY
STATE                                              PINCODE    REGISTRAT


10 rows selected.
```

Search for a particular customer

```
SQL> SELECT Customer_ID, First_Name, Last_Name, Email, Phone
  2  FROM Customer
  3  WHERE Customer_ID = 1;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
----------
          1 Aarav
Mehta
aarav.mehta@gmail.com
9000000001
```

This displays the details of the customer whose ID is 1.

4. Apply Filtering Conditions

Multiple conditions can be combined using operators such as AND, OR, and IN.

4.1 AND Condition

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price > 200
  4  AND Category_ID = 1;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1002
Rose Face Cream
       399

      1001
Aloe Vera Face Wash
       299

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
```

This displays products priced above ₹200 that belong to Category ID 1.

4.2 OR Condition

```
SQL> SELECT Product_ID, Product_Name, Category_ID, Price
  2  FROM Product
  3  WHERE Category_ID = 1
  4  OR Category_ID = 2;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE
----------- ----------
      1002
Rose Face Cream
          1        399

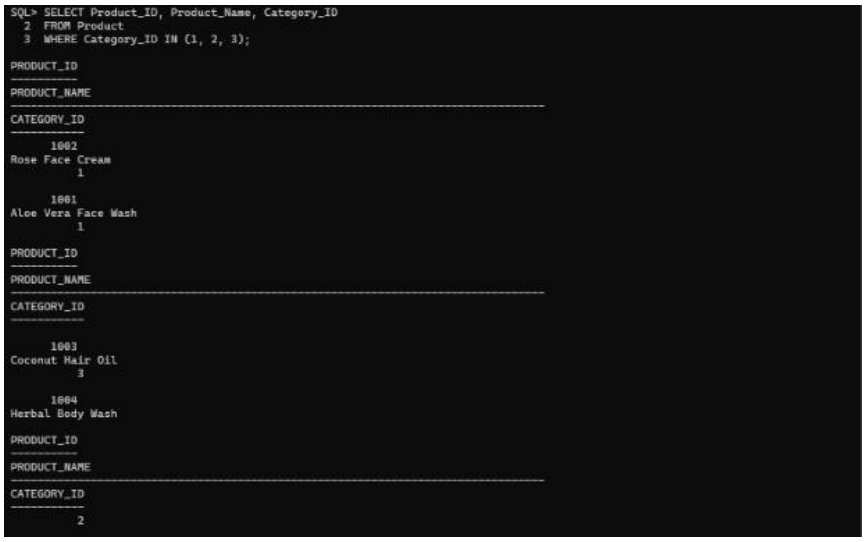
      1001
Aloe Vera Face Wash
          1        299

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE
----------- ----------

      1004
Herbal Body Wash
          2        449
```

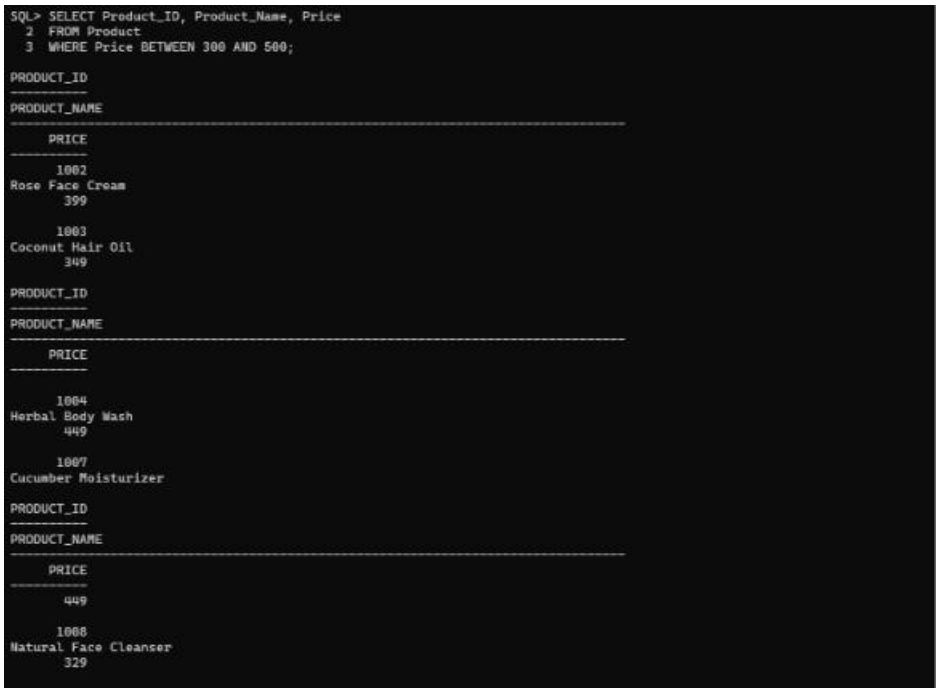
This displays products belonging to either Category ID 1 or Category ID 2.

4.3 IN Operator



This displays products belonging to any of the specified categories.

4.4 BETWEEN Operator



This displays products priced between ₹300 and ₹500.

5. Basic Business Reports

5.1 Product Price Report



```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1004
Herbal Body Wash
       449

      1007
Cucumber Moisturizer
       449

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

      1002
Rose Face Cream
       399

      1003
Coconut Hair Oil

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       349

      1008
Natural Face Cleanser
       329

      1001

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Aloe Vera Face Wash
       299

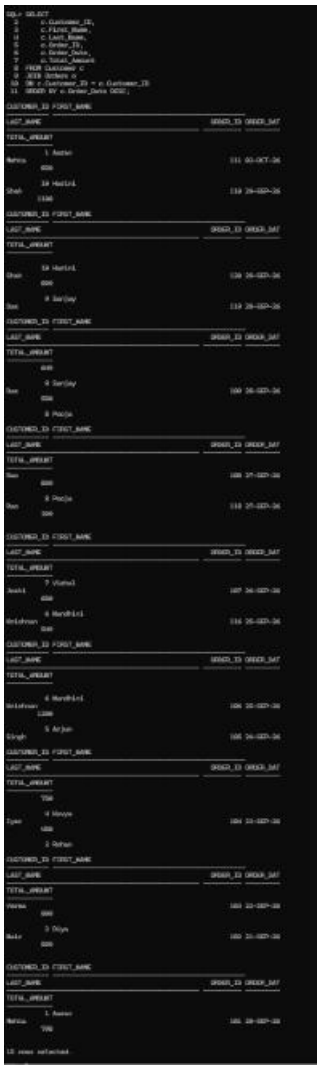
      1006
Neem Face Pack
       279


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1005
Natural Lip Balm
       199


8 rows selected.
```

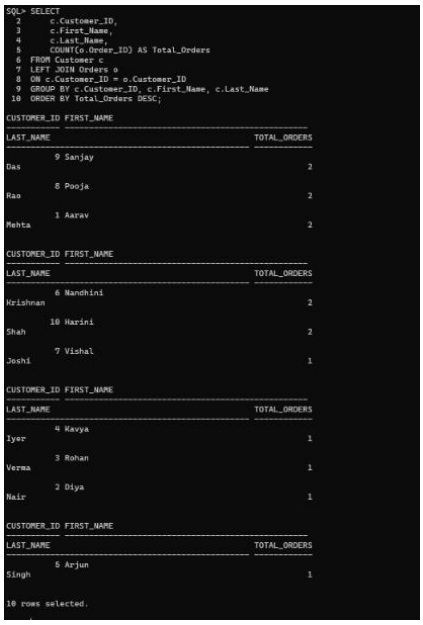
This report displays products sorted from the highest price to the lowest price.

5.2 Customer Order Report



This report displays customer information along with their order details.

5.3 Customer-wise Total Orders



```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name,
  4      c.Last_Name,
  5      COUNT(o.Order_ID) AS Total_Orders
  6  FROM Customer c
  7  LEFT JOIN Orders o
  8  ON c.Customer_ID = o.Customer_ID
  9  GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
 10  ORDER BY Total_Orders DESC;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS
-------------------------------------------------- ------------
          9 Sanjay
Das                                                           2

          8 Pooja
Rao                                                           2

          1 Aarav
Mehta                                                         2

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS
-------------------------------------------------- ------------
          6 Nandhini
Krishnan                                                      2

         10 Harini
Shah                                                          2

          7 Vishal
Joshi                                                         1

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS
-------------------------------------------------- ------------
          4 Kavya
Iyer                                                          1

          3 Rohan
Verma                                                         1

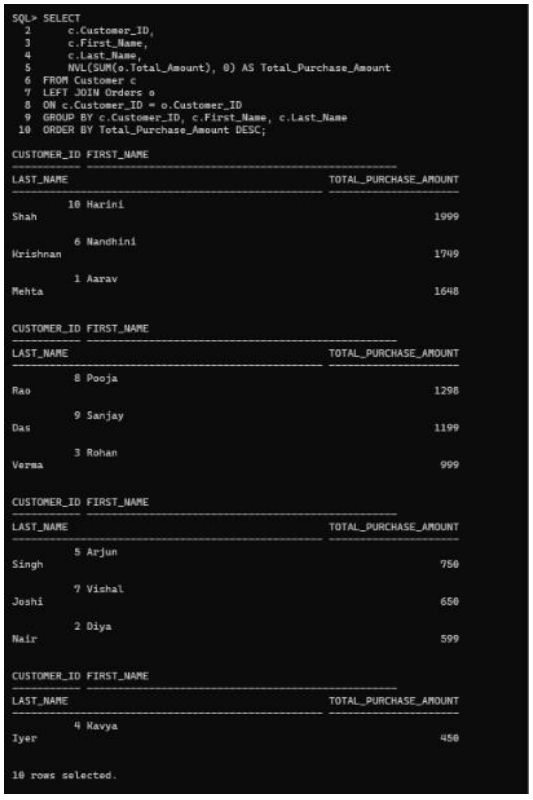
          2 Diya
Nair                                                          1

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS
-------------------------------------------------- ------------
          5 Arjun
Singh                                                         1

10 rows selected.
```

This report calculates the total number of orders placed by each customer, including customers who have not placed any orders.

5.4 Customer-wise Total Purchase Amount



```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name,
  4      c.Last_Name,
  5      NVL(SUM(o.Total_Amount), 0) AS Total_Purchase_Amount
  6  FROM Customer c
  7  LEFT JOIN Orders o
  8  ON c.Customer_ID = o.Customer_ID
  9  GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
 10  ORDER BY Total_Purchase_Amount DESC;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_PURCHASE_AMOUNT
-------------------------------------------------- ---------------------
         10 Harini
Shah                                                                1999

          6 Nandhini
Krishnan                                                            1749

          1 Aarav
Mehta                                                               1648

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_PURCHASE_AMOUNT
-------------------------------------------------- ---------------------
          8 Pooja
Rao                                                                 1298

          9 Sanjay
Das                                                                 1199

          3 Rohan
Verma                                                                999

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_PURCHASE_AMOUNT
-------------------------------------------------- ---------------------
          5 Arjun
Singh                                                                750

          7 Vishal
Joshi                                                                650

          2 Diya
Nair                                                                 599

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_PURCHASE_AMOUNT
-------------------------------------------------- ---------------------
          4 Kavya
Iyer                                                                 450

10 rows selected.
```

This report calculates the total order amount for each customer. NVL() displays zero for customers without orders.